UNIT 03

Forces and Motion

Newton’s laws, free-body diagrams, friction, and connected objects.



3.1

Newton’s Laws

How forces affect motion.



What Is a Force?

A force is a push or a pull.



A vector

A force has a size and a direction.

Measured in newtons (N)

1 N is about the weight of a small apple.

The direction matters just as much as the number.

Mass and Weight

Mass measures how much matter an object has. Weight is the force of gravity.

Mass

Measured in kilograms (kg).
Stays the same when you travel.

Weight

Measured in newtons (N).
Changes when gravity changes.

10 kg on Earth

$F_g = 10(9.8) = 98$ N

10 kg on the Moon

$F_g = 10(1.6) = 16$ N

Weight: $F_g = mg$. On Earth, $g = 9.8$ m/s².

Weight on Earth and the Moon

A student has a mass of 60 kg. Find their weight in each location.

SAME MASS

60 kg

Different gravity.
Different weight.

EQUATION

$F_g = mg$

EARTH: g = 9.8 m/s²

$F_g = 60(9.8) = 588$ N down

MOON: g = 1.6 m/s²

$F_g = 60(1.6) = 96$ N down



Newton’s First Law

An object at rest stays at rest, and an object in motion keeps a constant velocity, unless a net external force acts on it.



Equilibrium

Forces balance: $\sum F = 0$.

Two possibilities

The object stays at rest, or moves at constant speed in a straight line.

A moving puck does not need a net forward force to keep moving.

Newton’s Second Law

An object’s acceleration is directly proportional to the net force and inversely proportional to its mass.

$$\sum F = ma$$

$$a = \sum F / m$$

In everyday words: same mass

More net force gives more acceleration.

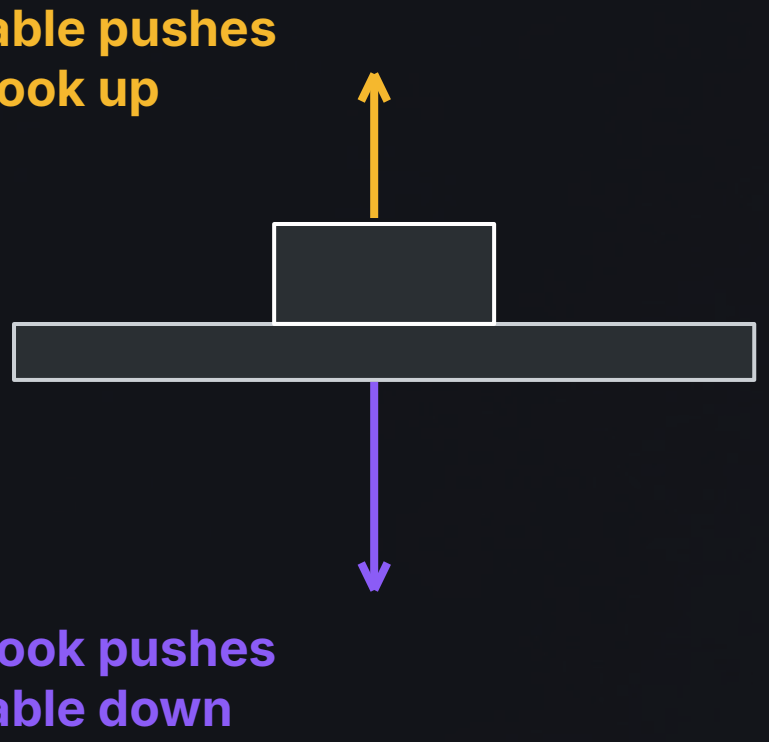
Same net force

Twice the mass gives half the acceleration.

Net force means the sum of all forces acting on the object.

Newton’s Third Law

For every action, there is an equal and opposite reaction.



The pair acts on two objects

Book on table. Table on book.

On the book alone

Gravity and the table’s push balance, but they are not a third-law pair.

Third-law forces do not cancel on a single object’s free-body diagram.

3.2

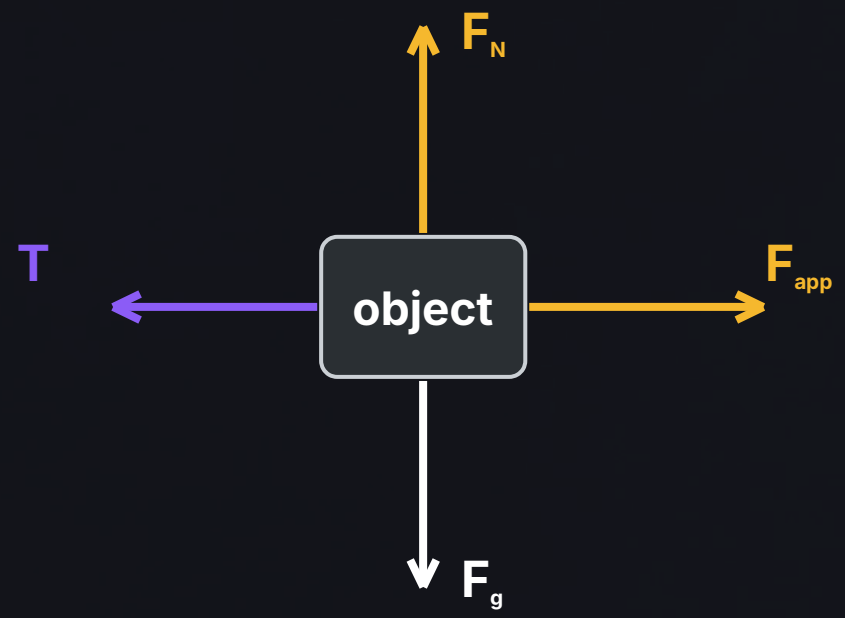
Free-Body Diagrams

One object. Every force acting on it.



Drawing a Free-Body Diagram

Use a box or dot for the object. Add one labeled arrow for each force.



Common forces

F_g: gravity pulls down
F_N: surface pushes perpendicular
T: rope pulls along the rope

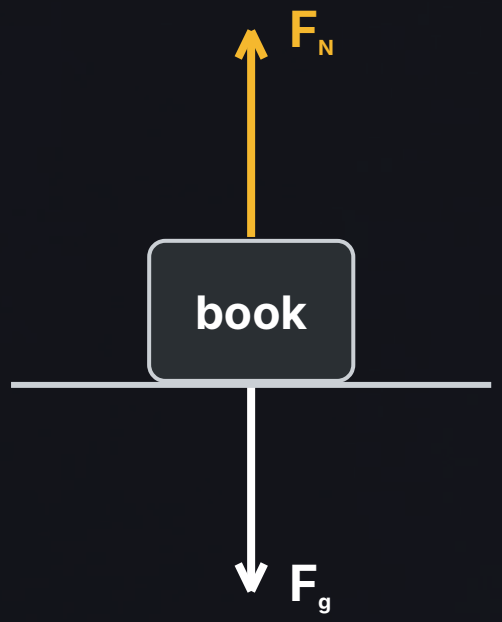
Arrow check

Longer arrows mean larger forces.
Keep motion arrows separate.

Show only the forces that act on the chosen object.

Normal Force

Normal means perpendicular. The normal force is a surface's push on an object.



What it does

A surface pushes on an object that presses against it.

Which way does it point?

Straight out from the surface. On a floor: up. On a ramp: tilted.

On a level surface with no other vertical forces or acceleration: $F_N = F_g$.

Balanced Forces: Book and Lamp

Both objects are at rest. Find the upward force on each.

BOOK ON A TABLE



$F_N = F_g = 2(9.8) = 19.6$ N

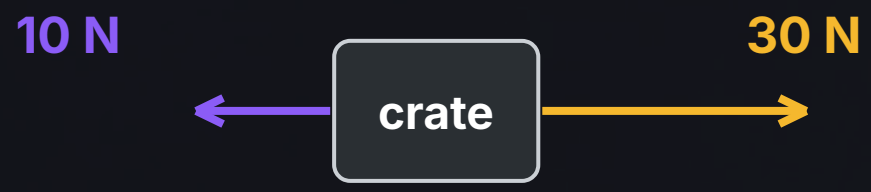
LAMP ON A ROPE



$T = F_g = 1.5(9.8) = 14.7$ N

Net Force in One Dimension

Choose a positive direction, then add forces with signs.



CHOOSE RIGHT AS POSITIVE

Right: +30 N Left: −10 N

ADD THE FORCES

$\sum F = 30 - 10 = 20$ N

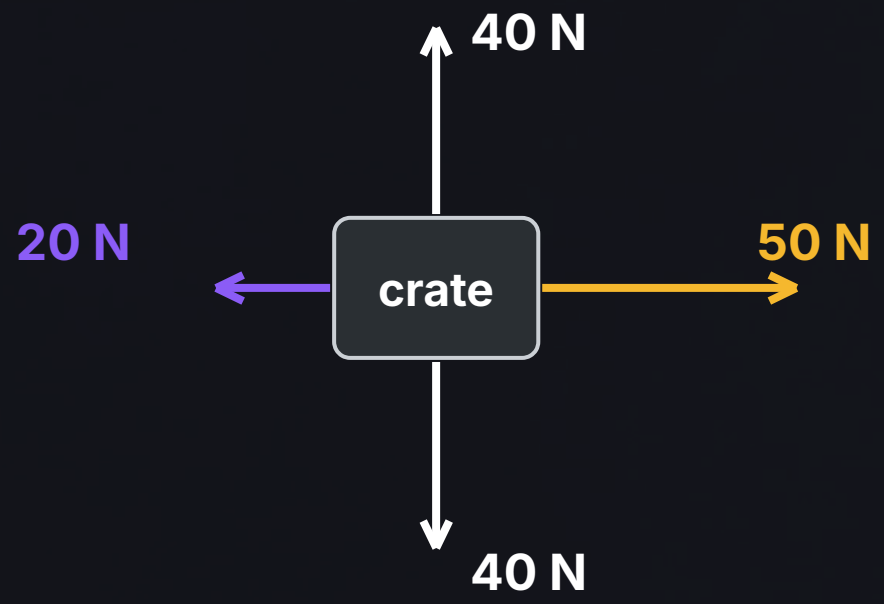
DIRECTION

20 N to the right

Add horizontal forces and vertical forces separately.

Equilibrium: What Force Is Missing?

A crate has forces of 50 N right, 20 N left, 40 N up, and 40 N down.



VERTICAL

$\sum F_y = 40 - 40 = 0$ N

HORIZONTAL

$\sum F_x = 50 - 20 = 30$ N right

FOR EQUILIBRIUM

Add 30 N to the left.

Resultant Force and Direction

Two perpendicular forces act on an object: 40 N right and 30 N up.



MAGNITUDE

$\sum F = \sqrt{(40^2 + 30^2)}$

RESULT

$\sum F = 50$ N

DIRECTION

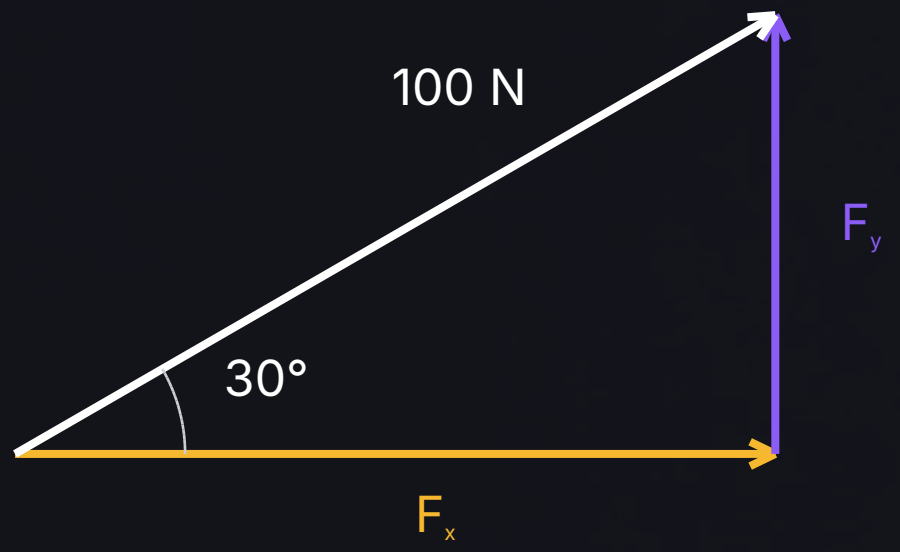
$\theta = \tan^{-1}(30/40) = 36.9°$

ANSWER

50 N, 36.9° above +x

Breaking a Force into Components

A 100 N force pulls 30° above the horizontal.



HORIZONTAL

$F_x = F \cos \theta = 100 \cos 30°$

VERTICAL

$F_y = F \sin \theta = 100 \sin 30°$

COMPONENTS

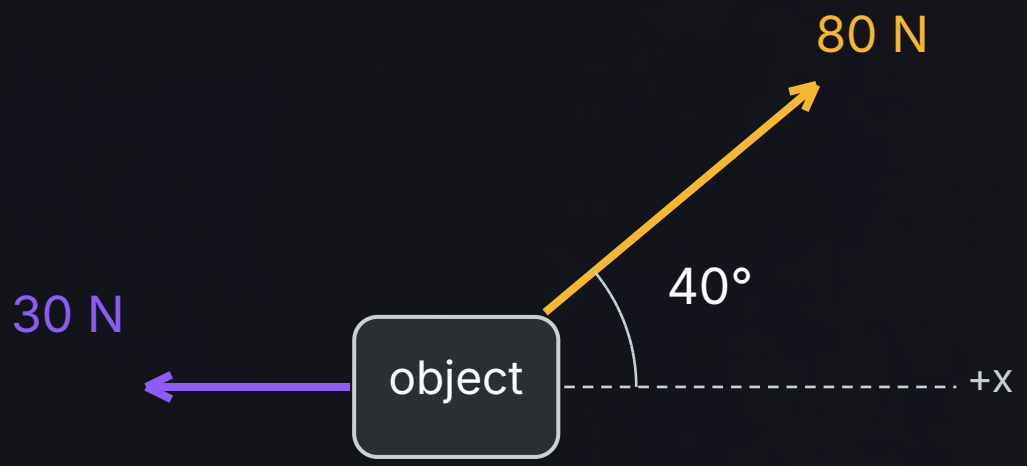
$F_x = 86.6$ N $F_y = 50.0$ N

CHECK

$\sqrt{(86.6^2 + 50.0^2)} \approx 100$ N

One Angled Force and One Straight Force

Top view: an 80 N force acts 40° above +x. A second force is 30 N left.



COMPONENTS

$F_x = 80 \cos 40° - 30 = 31.3 \text{ N}$

PERPENDICULAR COMPONENT

$F_y = 80 \sin 40° = 51.4 \text{ N}$

MAGNITUDE

$\sum F = \sqrt{(31.3^2 + 51.4^2)} \approx 60 \text{ N}$

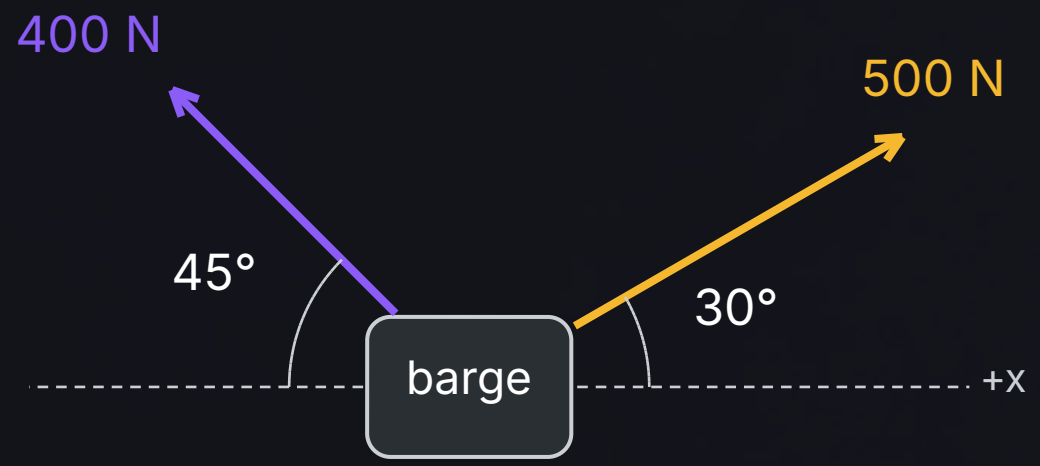
DIRECTION

$\theta = \tan^{-1}(51.4/31.3) \approx 59°$

Here, +y is up the page. Weight and normal force balance out of the page.

Two Angled Forces: Tugboats

One boat pulls 500 N at 30° above +x. Another pulls 400 N at 45° above −x.



HORIZONTAL

$F_x = 500 \cos 30° - 400 \cos 45°$
$= 150$ N

VERTICAL

$F_y = 500 \sin 30° + 400 \sin 45°$
$= 533$ N

MAGNITUDE

$\sum F = \sqrt{(150^2 + 533^2)} \approx 554$ N

DIRECTION

$\theta = \tan^{-1}(533/150) \approx 74°$

Top view. Check the quadrant before reporting an angle.

3.3

Forces on an Incline

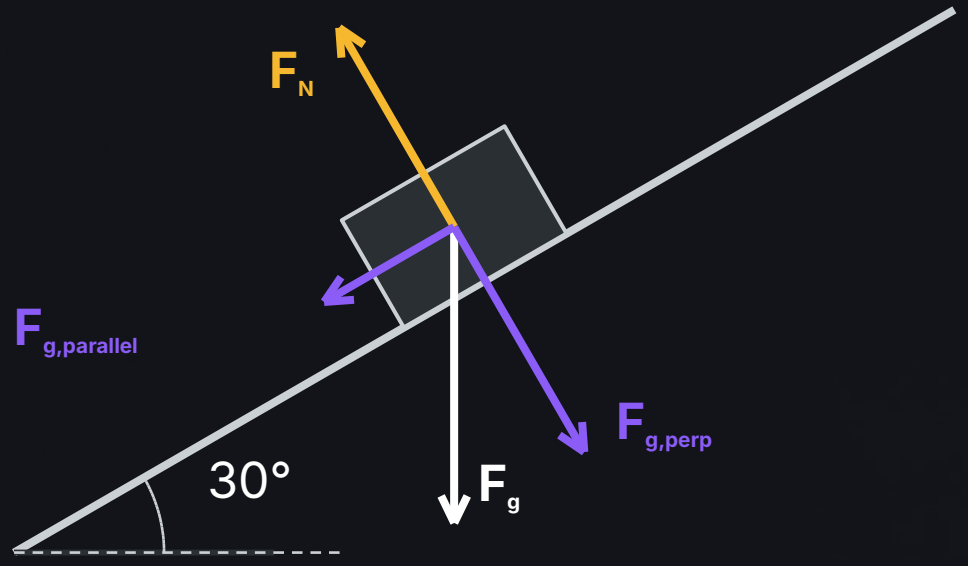
Tilt the axes to match the slope.



Gravity on a Ramp

Use one axis along the ramp and one perpendicular to it.

Purple arrows split F_g into parts.



Along the ramp

$F_{g,parallel} = mg \sin \theta$

Points down the slope.

Into the ramp

$F_{g,perp} = mg \cos \theta$

Points into the surface.

θ is also the angle between weight and the direction straight into the ramp.

Normal Force on an Incline

The normal force pushes perpendicular to the surface.

$$F_N = mg \cos \theta$$

15° slope
$F_N \approx 0.97mg$

60° slope
$F_N = 0.50mg$

Nearly flat: $F_N \approx mg$. Nearly vertical: F_N approaches zero.

Why it gets smaller

A steeper ramp supports less of the weight perpendicular to the surface.

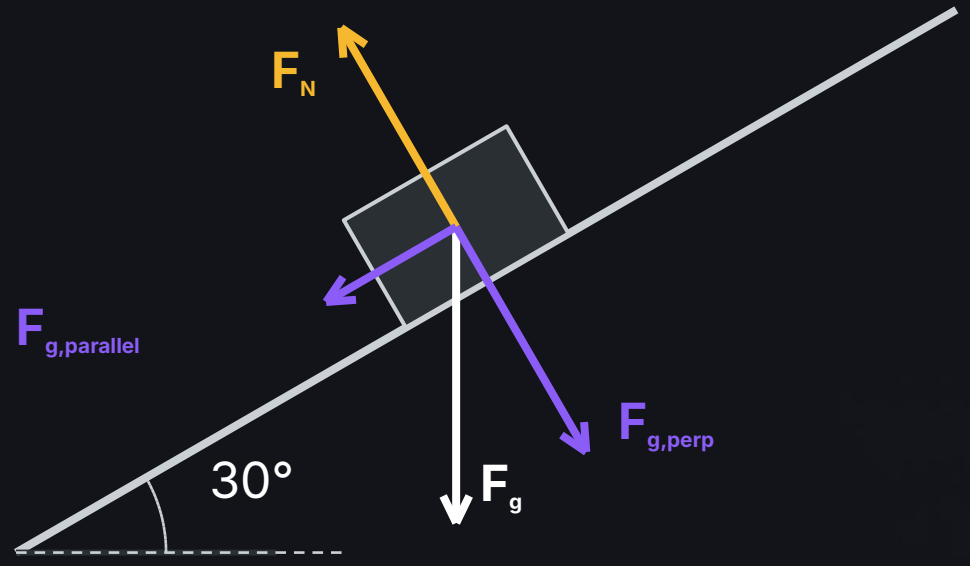
When this equation works

No extra perpendicular force and no perpendicular acceleration.

Incline Example: A 5 kg Block

A 5 kg block slides on a frictionless 30° ramp. Find the weight components and normal force.

Purple arrows split F_g into parts.



WEIGHT

$F_g = mg = 5(9.8) = 49$ N

DOWN THE RAMP

$F_{g,parallel} = 49 \sin 30° = 24.5$ N

INTO THE RAMP

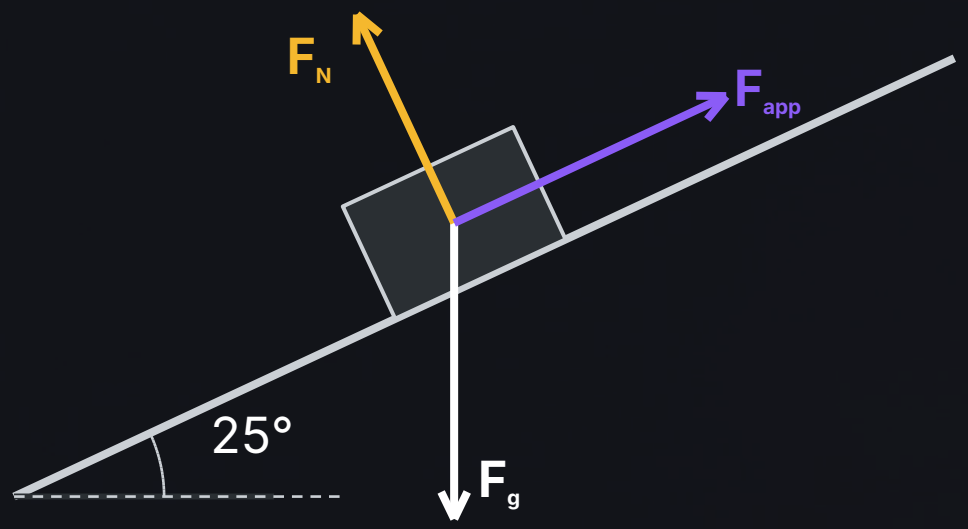
$F_{g,perp} = 49 \cos 30° = 42.4$ N

NORMAL FORCE

$\mathbf{F_N = 42.4\ N}$

Incline Example: A Push Up the Ramp

A 12 kg block is on a frictionless 25° ramp. A 30 N force pushes it uphill.



CHOOSE DOWNHILL AS POSITIVE

$F_{g,parallel} = 12(9.8) \sin 25°$

GRAVITY COMPONENT

$F_{g,parallel} = 49.7$ N downhill

NET FORCE

$\sum F = 49.7 - 30 = 19.7$ N

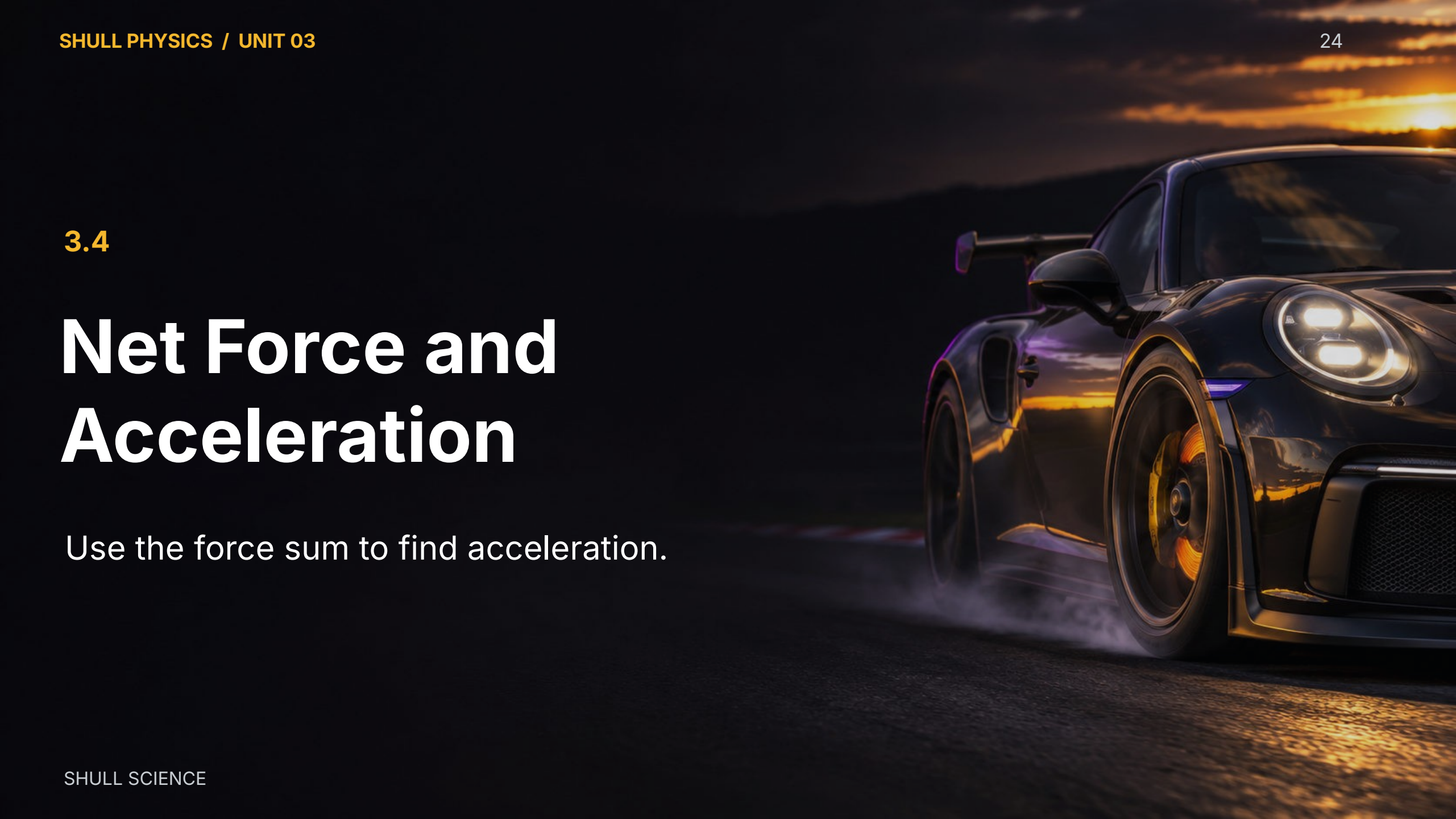
DIRECTION

19.7 N down the ramp

3.4

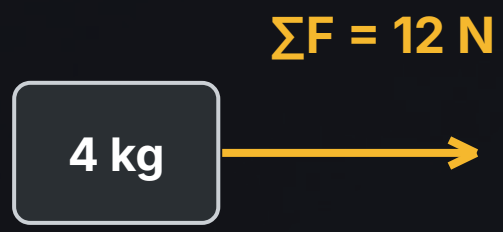
Net Force and Acceleration

Use the force sum to find acceleration.



Calculating Acceleration

A 4 kg object has a net force of 12 N to the right.



EQUATION

$\sum F = ma$

SOLVE FOR a

$a = \sum F / m$

SUBSTITUTE

$a = 12\ N / 4\ kg$

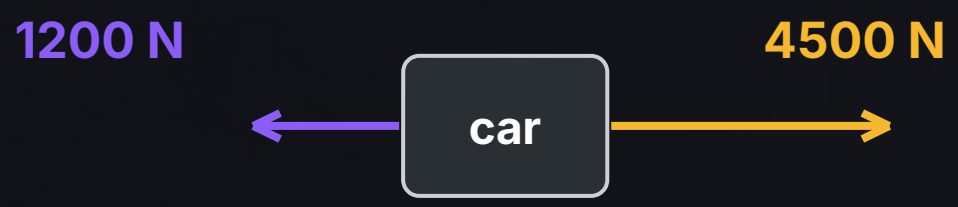
ANSWER

$a = 3.0\ m/s^2$ right

1 N = 1 kg·m/s². Acceleration points in the direction of the net force.

A Car Accelerates

A 1500 kg car has 4500 N forward force and 1200 N of drag.



CHOOSE FORWARD AS POSITIVE

$\sum F = 4500 - 1200$

NET FORCE

$\sum F = 3300$ N forward

ACCELERATION

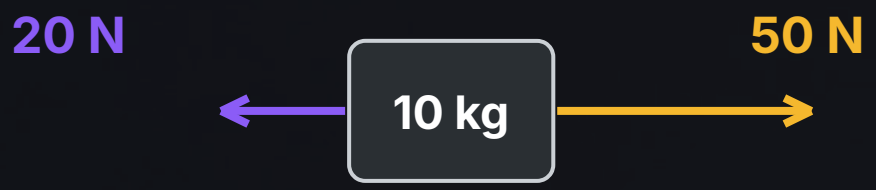
$a = \sum F / m = 3300 / 1500$

ANSWER

$a = 2.2$ m/s² forward

From a Diagram to Acceleration

A 10 kg cart has a 50 N push right and a 20 N opposing force left.



CHOOSE RIGHT AS POSITIVE

$\sum F = 50 - 20$

NET FORCE

$\sum F = 30$ N right

NEWTON'S SECOND LAW

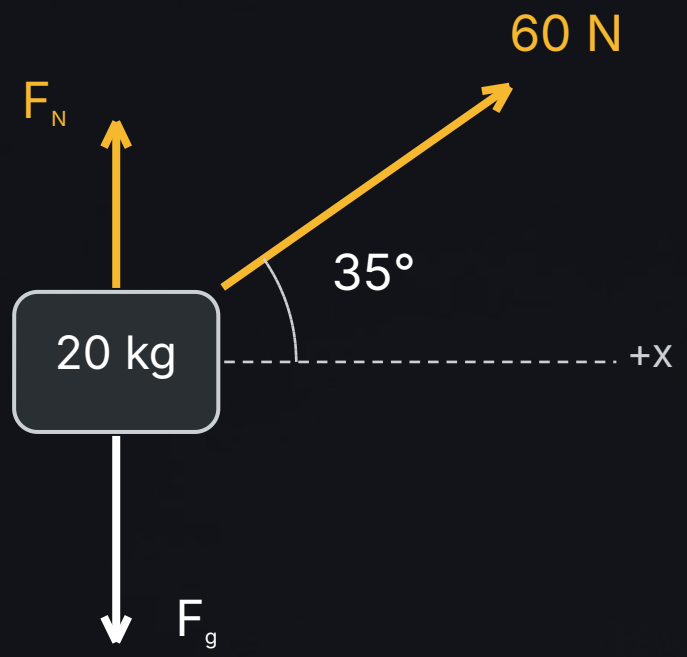
$a = \sum F / m = 30 / 10$

ANSWER

$a = 3.0$ m/s² right

A Sled Pulled at an Angle

A 20 kg sled is pulled with 60 N at 35° above horizontal. Ignore friction.



HORIZONTAL FORCE

$F_x = 60 \cos 35° = 49.1$ N

ACCELERATION

$a = F_x / m = 49.1 / 20$

$= 2.46$ m/s² right

VERTICAL BALANCE

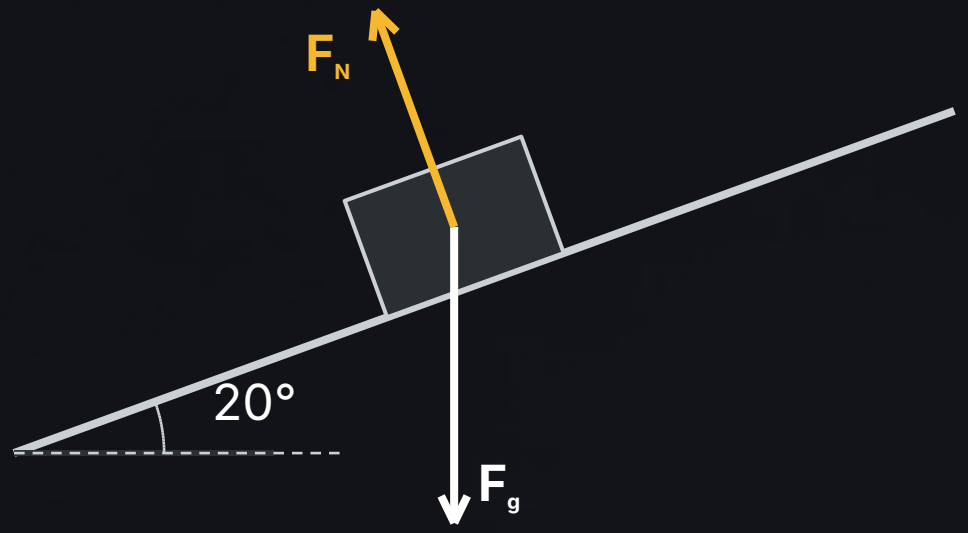
$F_N + 60 \sin 35° - mg = 0$

NORMAL FORCE

$F_N = 196 - 34.4 \approx 162$ N

Acceleration Down a Frictionless Ramp

An 8 kg block slides down a 20° ramp. Find its acceleration.



FORCE ALONG THE RAMP

$$\sum F = mg \sin \theta$$

NEWTON'S SECOND LAW

$$ma = mg \sin \theta$$

MASS CANCELS

$$a = g \sin \theta = 9.8 \sin 20°$$

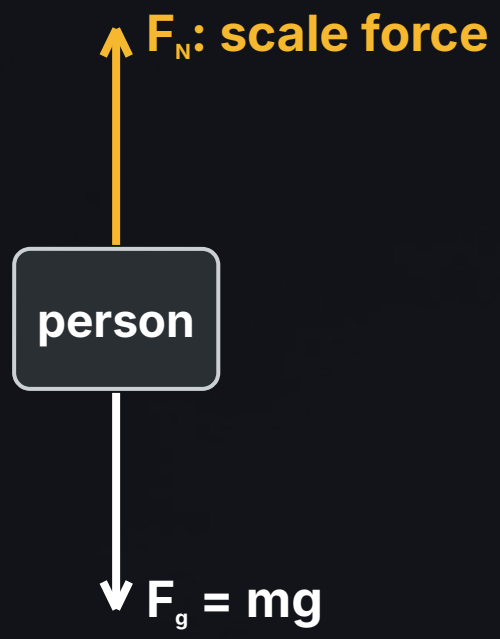
ANSWER

$a = 3.35\ m/s^2$ downhill

Without friction, the acceleration does not depend on the block's mass.

Elevators and Apparent Weight

A scale reads the upward normal force, F_N. That is your apparent weight.



Choose up as positive

$F_N - mg = ma$

Rest or constant velocity

$a = 0$, so $F_N = mg$.
The scale reads your usual weight.

The scale reading changes with acceleration, not simply with motion up or down.

Elevator Accelerating Up

A 70 kg student accelerates upward at 2.0 m/s². What does the scale read?



WEIGHT

$mg = 70(9.8) = 686$ N

UP IS POSITIVE

$F_N - mg = ma$

SUBSTITUTE

$F_N = 686 + 70(2.0)$

SCALE READING

$F_N = 826$ N

Elevator Accelerating Down

The same 70 kg student accelerates downward at 2.0 m/s².



UP IS POSITIVE

$a = -2.0\ m/s^2$

EQUATION

$F_N - mg = ma$

SUBSTITUTE

$F_N = 686 + 70(-2.0)$

SCALE READING

$F_N = 546$ N

3.5

Friction

What changes when surfaces resist sliding?



Static and Kinetic Friction

Friction opposes sliding, or the tendency to slide, between surfaces.

Static: surfaces are not sliding

f_s adjusts to match the needed force.
Maximum: $f_{s,max} = \mu_s F_N$

Kinetic: surfaces are sliding

$f_k \approx \mu_k F_N$
Usually smaller than maximum static.

At rest with a 20 N push:
f_s = 20 N

Keep sliding → use μ_k

μ has no units. In this model, kinetic friction stays about constant when μ_k and F_N stay constant.

Comparing Coefficients of Friction

At the same normal force: higher μ_k means more sliding friction.

Surfaces in contact	Static μ_s	Sliding μ_k
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.7	0.5
Waxed wood on wet snow	0.14	0.10
Shoes on ice	0.10	0.05
Teflon on steel	0.04	0.04
Synovial joints (lubricated joint surfaces)	0.016	0.015

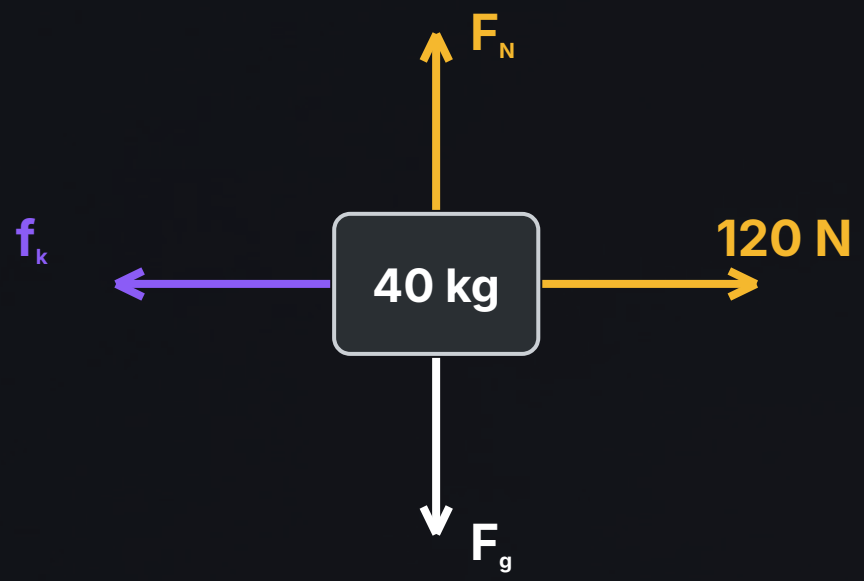
Your knee can slide with less friction than shoes on ice.

Approximate values. μ has no units. Conditions such as moisture, wear, and temperature matter.

Source: OpenStax, College Physics 2e, Table 5.1

Friction at Constant Velocity

A 40 kg box moves across a level floor at constant velocity under a horizontal 120 N push.



VERTICAL BALANCE

$F_N = mg = 40(9.8) = 392$ N

CONSTANT VELOCITY

$a = 0$, so $f_k = 120$ N

FRICTION EQUATION

$\mu_k = f_k / F_N$

ANSWER

$\mu_k = 120 / 392 \approx 0.31$

Will the Box Move?

A 25 kg box starts at rest. $\mu_s = 0.40$ and $\mu_k = 0.30$. Compare 90 N and 110 N pushes.

$F_N = 25(9.8) = 245$ N

$f_{s,max} = 0.40(245) = 98$ N

90 N push

90 N < 98 N
Static friction is 90 N.
The box stays at rest.

110 N PUSH

110 N > 98 N: the box slides.

KINETIC FRICTION

$f_k = 0.30(245) = 73.5$ N

NET FORCE

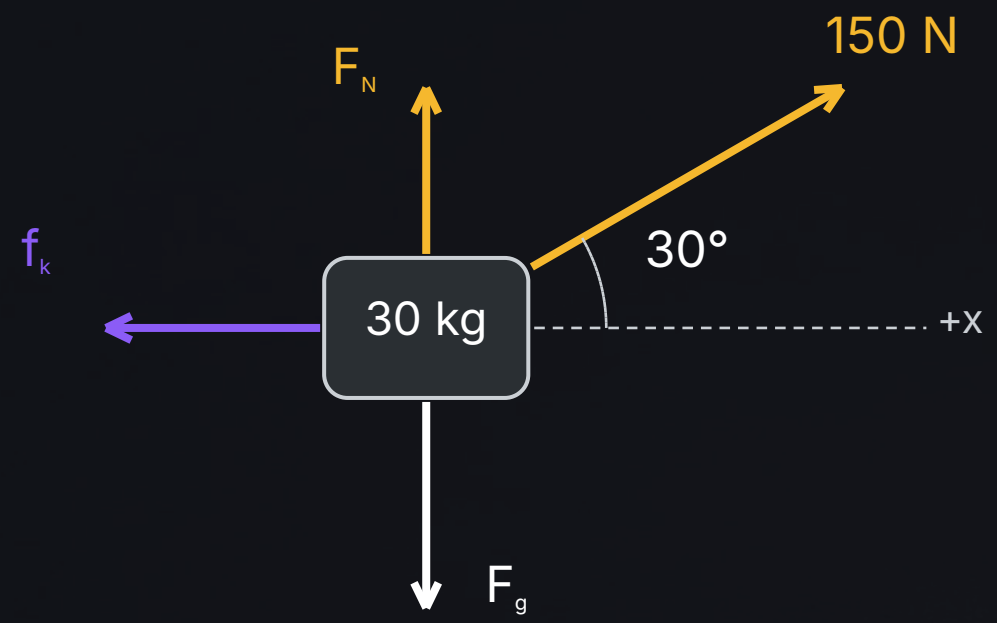
$\sum F = 110 - 73.5 = 36.5$ N

ACCELERATION

$a = 36.5 / 25 = 1.46$ m/s²

Angled Pull with Friction

A 150 N rope pulls a 30 kg box at 30° above horizontal. The box is sliding. $\mu_k = 0.25$.



NORMAL FORCE

$F_N = mg - F \sin 30°$
$= 294 - 75 = 219$ N

FRICTION

$f_k = 0.25(219) = 54.75$ N

HORIZONTAL NET FORCE

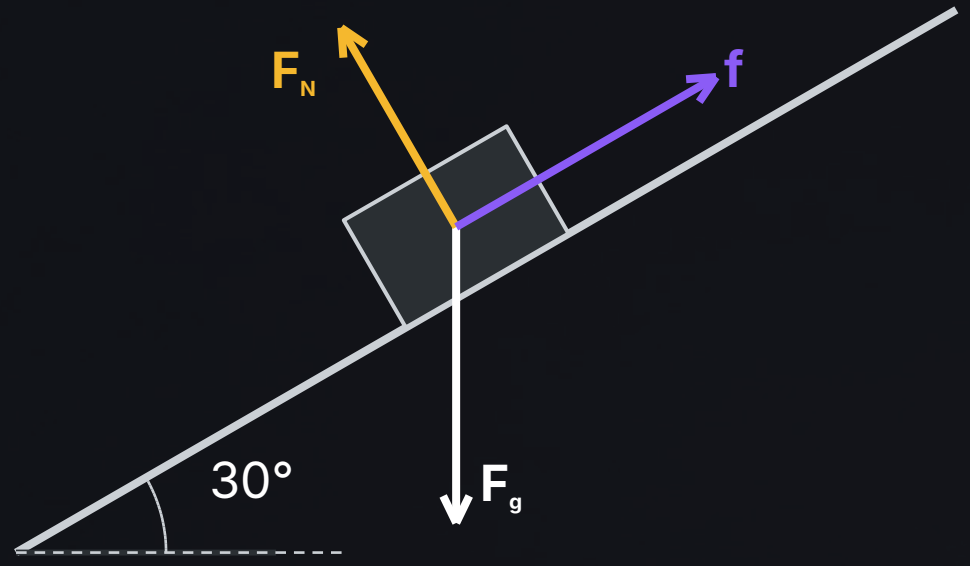
$\sum F = 150 \cos 30° - 54.75$
$= 75.15$ N

ACCELERATION

$a = 75.15 / 30 = 2.51$ m/s²

Friction on an Incline

For a block sliding down a ramp, friction points up the ramp.



Normal and friction

$F_N = mg \cos \theta$

Maximum static: $f_{s,max} = \mu_s F_N$

Sliding: $f_k \approx \mu_k F_N$

Will it start sliding down?

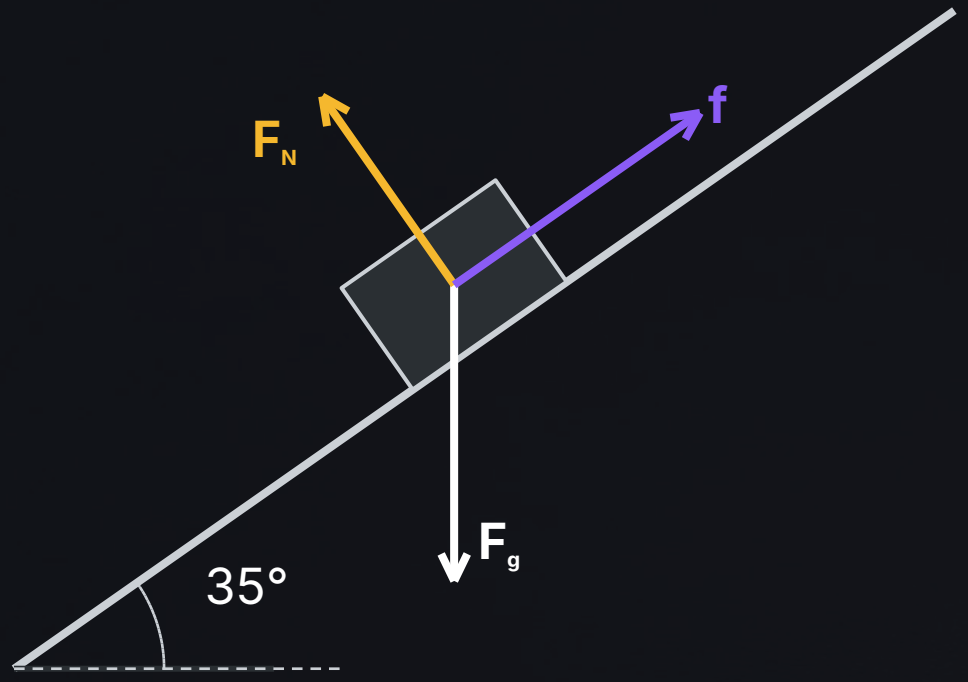
Compare $mg \sin \theta$ with $\mu_s F_N$.

It slides if $mg \sin \theta > \mu_s F_N$.

These equations assume gravity and contact forces only. Friction reverses if sliding reverses.

Will It Slide Down the Ramp?

A 10 kg block starts at rest on a 35° ramp. $\mu_s = 0.50$ and $\mu_k = 0.40$.



GRAVITY AND NORMAL

$F_{g,parallel} = 10(9.8) \sin 35° = 56.2$ N

$F_N = 10(9.8) \cos 35° = 80.3$ N

STATIC LIMIT

$f_{s,max} = 0.50(80.3) = 40.1$ N

$56.2 > 40.1$, so it slides.

SLIDING FRICTION

$f_k = 0.40(80.3) = 32.1$ N

ACCELERATION

$a = (56.2 - 32.1)/10$

$= 2.41$ m/s² downhill

3.6

Connected Objects

One rope connects the motion.



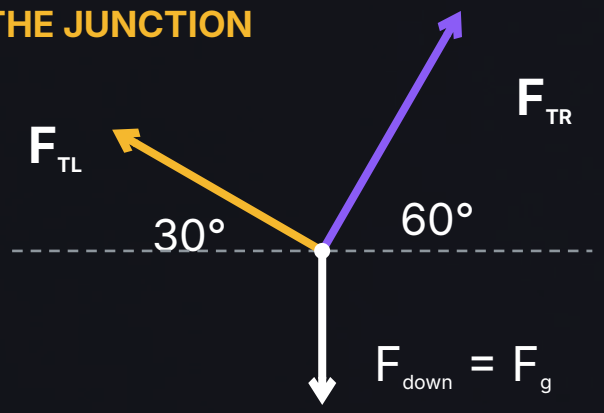
Streetlight Cable Tension

A 20 kg streetlight hangs at rest. Find the tension in each support cable.

Left cable: 30° above horizontal. Right cable: 60°. Ignore cable mass.



FORCES AT THE JUNCTION



Angles measured from horizontal. Arrows not to scale.

1 AT REST: FORCES BALANCE

$\sum F_x = 0 \quad \sum F_y = 0$

$F_{down} = mg = 20(9.8) = 196\ N$

2 HORIZONTAL BALANCE

$F_{TR} \cos 60° = F_{TL} \cos 30°$

$F_{TL} = 0.577 F_{TR}$

3 VERTICAL BALANCE

$F_{TL} \sin 30° + F_{TR} \sin 60° = 196$

$(0.577F_{TR})(0.5) + 0.866F_{TR} = 196$

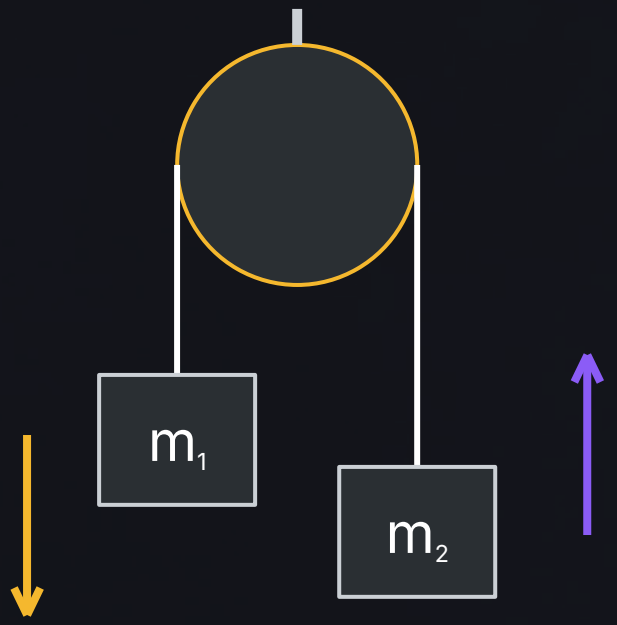
4 TENSION IN EACH CABLE

$F_{TR} \approx 170\ N \quad F_{TL} = 98.0\ N$

The steeper cable carries more tension.

Ideal Strings and Pulleys

Use a massless, taut rope that does not stretch, and a massless, frictionless pulley.



Motion directions

Draw a separate free-body diagram for each object.

For these fixed-pulley systems

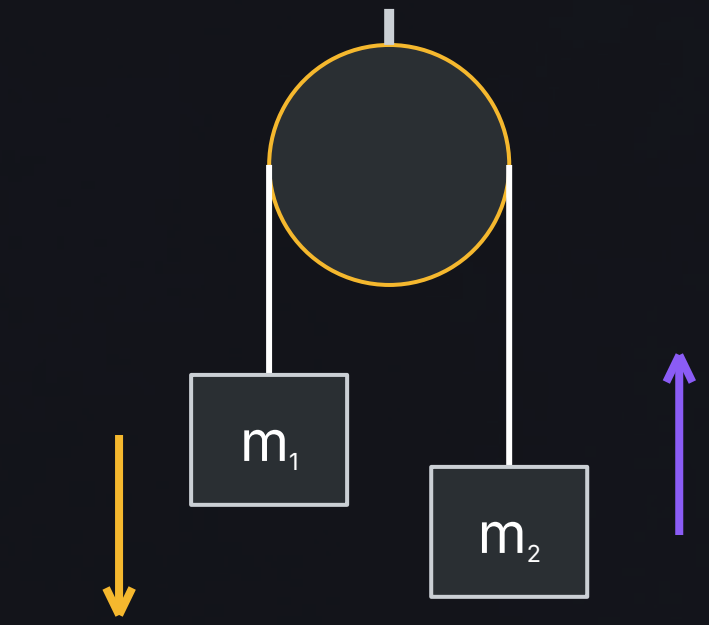
The rope has the same tension, T. Both objects have the same acceleration size.

A useful approach

Find a using the whole system. Then use one object to find T.

Atwood Machine

Two hanging masses share a rope: m_1 = 5 kg and m_2 = 3 kg. Find acceleration and tension.



5 kg DOWN, 3 kg UP

$\sum F = (5 - 3)(9.8) = 19.6$ N

WHOLE SYSTEM

$a = 19.6 / (5 + 3) = 2.45$ m/s²

USE THE 3 kg MASS

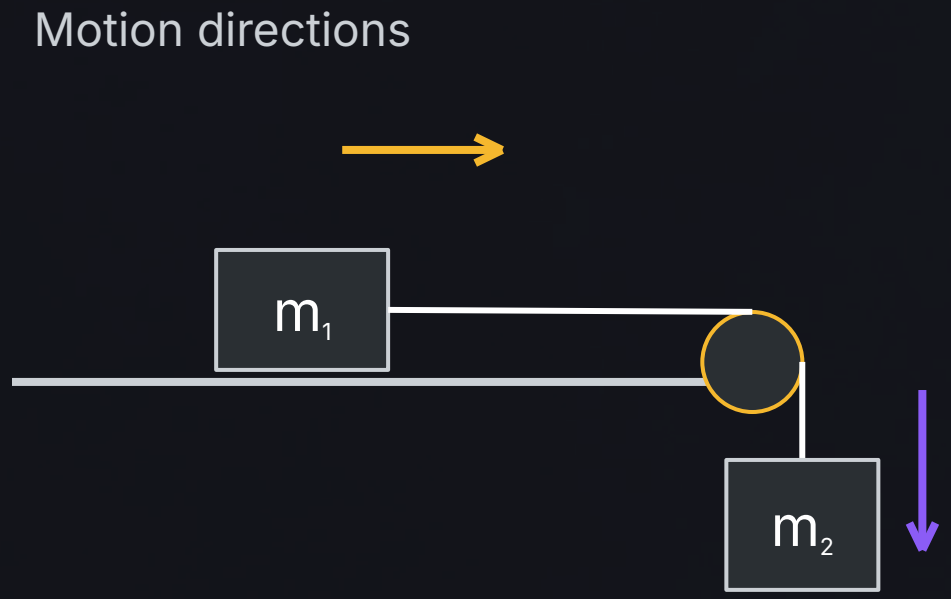
$T - 3g = 3a$

TENSION

$T = 3(9.8 + 2.45) = 36.8$ N

A Block on a Frictionless Table

A 4 kg block on a table connects to a hanging 2 kg mass.



DRIVING FORCE

$\sum F = m_2 g = 2(9.8) = 19.6$ N

WHOLE SYSTEM

$a = 19.6 / (4 + 2) = 3.27$ m/s²

TABLE BLOCK

$T = m_1 a$

TENSION

$T = 4(3.27) \approx 13.1$ N

The hanging mass accelerates down. The table block accelerates toward the pulley.

The Same System with Friction

The 4 kg block slides on the table with $\mu_k = 0.20$. The hanging mass is still 2 kg.

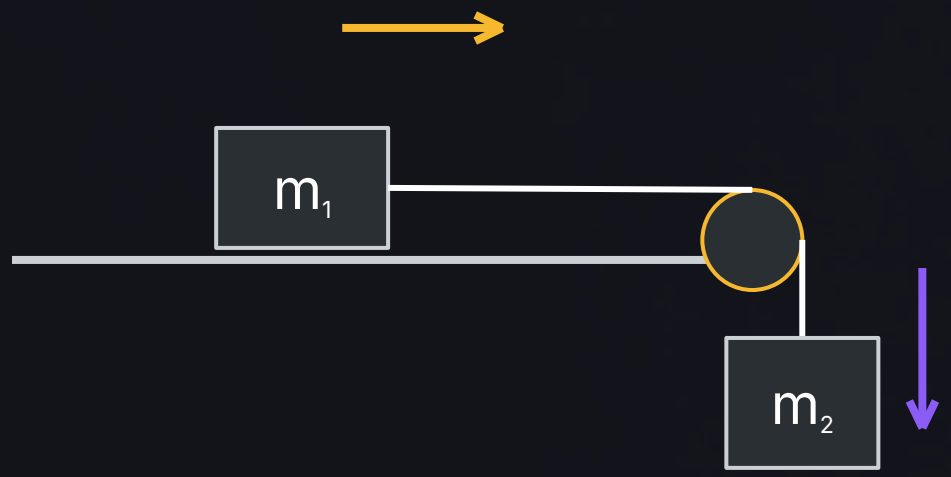


TABLE FRICTION

$$f_k = \mu_k m_1 g = 0.20(4)(9.8) = 7.84 \text{ N}$$

WHOLE SYSTEM

$$a = (19.6 - 7.84) / 6$$
$$= 1.96 \text{ m/s}^2$$

TABLE BLOCK

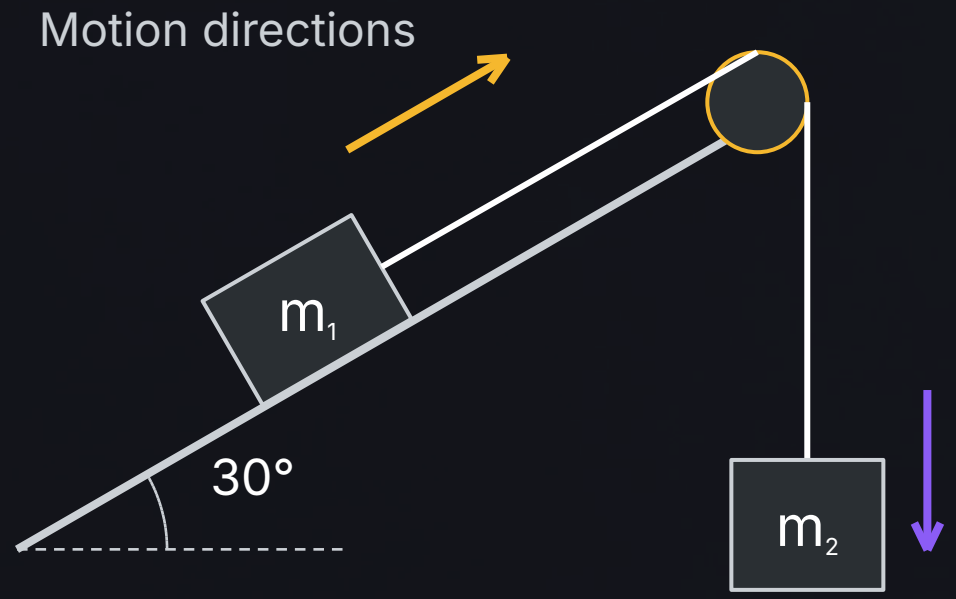
$$T - f_k = m_1 a$$

TENSION

$$\mathbf{T = 4(1.96) + 7.84 = 15.7 \text{ N}}$$

Challenge: Incline and Pulley

A 6 kg block on a frictionless 30° ramp connects to a hanging 5 kg mass.



OPPOSING WEIGHTS

Hanging: $5(9.8) = 49.0$ N

Along ramp: $6(9.8)\sin 30° = 29.4$ N

WHOLE SYSTEM

$a = (49.0 - 29.4) / (5 + 6)$

$= 1.78\ m/s^2$

HANGING MASS

$5g - T = 5a$

TENSION

$$\mathbf{T = 5(9.8 - 1.78) \approx 40.1\ N}$$

The 5 kg mass moves down while the 6 kg block moves up the ramp.